Lecture 3: Linear Classifiers

Part 2: SVM loss and regularization

Errahel Nazih

September 2026

Multiclass SVM Loss





cat	**3.2**	1.3	2.2
car	5.1	**4.9**	2.5
frog	-1.7	2.0	**-3.1**

Loss

Given an example (x_i, y_i)
(x_i is image, y_i is label)

Let $s = f(x_i, W)$ be scores

Then the SVM loss has the form:

$$L_i = \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + 1)$$

Multiclass SVM Loss





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Loss	2.9		

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Then the SVM loss has the form:

$$L_i = \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + 1)$$

= max(0, 5.1 -3.2 + 1)
+max(0, -1.7 -3.2 + 1)
= max(0, 2.9) + max(0, -3.9)
= 2.9 + 0 = 2.9

Multiclass SVM Loss





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Loss	2.9	0	

Given an example (x_i, y_i)
(x_i is image, y_i is label)

Let $s = f(x_i, W)$ be scores

Then the SVM loss has the form:

$$L_i = \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + 1)$$

= max(0, 1.3 -4.9 + 1)

+max(0, 2.0 -4.9 + 1)
= max(0, -2.6) + max(0, -1.9)

= 0 + 0
= 0

Multiclass SVM Loss





cat	**3.2**	1.3	2.2
car	5.1	**4.9**	2.5
frog	-1.7	2.0	**-3.1**
Loss	2.9	0	12.9

Given an example (x_i, y_i)
(x_i is image, y_i is label)

Let $s = f(x_i, W)$ be scores

Then the SVM loss has the form:

$$L_i = \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + 1)$$

= max(0, 2.2 -(-3.1) + 1)
+max(0, 2.5 -(-3.1) + 1) =
max(0, 6.3) + max(0, 6.6) =
6.3 + 6.6 = 12.9

Multiclass SVM Loss





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Given an example (x_i, y_i)
(x_i is image, y_i is label)

Let $s = f(x_i, W)$ be scores

Then the SVM loss has the form:

$$L_i = \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + 1)$$

Loss over the dataset is: L =

(2.9 + 0.0 + 12.9) / 3

= 5.27

Multiclass SVM Loss





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Given an example (x_i, y_i)
(x_i is image, y_i is label)

Let $s = f(x_i, W)$ be scores

Then the SVM loss has the form:

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Q1: What happens to the loss if the scores for the car image change slightly?

Multiclass SVM Loss





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Given an example (x_i, y_i)
(x_i is image, y_i is label)

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Q2: What are the minimum and maximum possible values of the loss?

Multiclass SVM Loss





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Given an example (x_i, y_i)
(x_i is image, y_i is label)

Let $s = f(x_i, W)$ be scores

Then the SVM loss has the form:

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Q3: If all class scores were random, what loss would we expect?

Multiclass SVM Loss





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Given an example (x_i, y_i)
(x_i is image, y_i is label)

Let $s = f(x_i, W)$ be scores

Then the SVM loss has the form:

$$L_i = \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + 1)$$

Q4: What would happen if the sum were over all classes? (including $i = y_i$)

Multiclass SVM Loss





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Given an example (x_i, y_i)
(x_i is image, y_i is label)

Let $s = f(x_i, W)$ be scores

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Q5: What changes if the loss uses a mean instead of a sum?

Multiclass SVM Loss





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Given an example (x_i, y_i)
(x_i is image, y_i is label)

Let $s = f(x_i, W)$ be scores

Then the SVM loss has the form:

$$L_i = \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + 1)$$

Q6: What if we used this loss instead?

$$L_i = \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + 1)^2$$

Multiclass SVM Loss

$$f(x, W) = Wx$$

$$L = \frac{1}{N} \sum_{i=1}^{N} \sum_{j \neq y_i} \max(0, f(x_i; W)_j - f(x_i; W)_{y_i} + 1)$$

Q: Suppose we found some W with L = 0. Is it unique?

Multiclass SVM Loss

$$f(x, W) = Wx$$

$$L = \frac{1}{N} \sum_{i=1}^{N} \sum_{j \neq y_i} \max(0, f(x_i; W)_j - f(x_i; W)_{y_i} + 1)$$

Q: Suppose we found some W with L = 0. Is it unique?

No. If W has zero loss, 2W can also have zero loss.

Multiclass SVM Loss

$$f(x, W) = Wx$$

$$L_i = \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + 1)^2$$





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Loss	2.9	0	12.9

Original W:

= max(0, 1.3 - 4.9 + 1)
+max(0, 2.0 - 4.9 + 1)
= max(0, -2.6) + max(0, -1.9)
= 0 + 0
= 0

Using 2W instead:

= max(0, 2.6 - 9.8 + 1)
+max(0, 4.0 - 9.8 + 1)
= max(0, -6.2) + max(0, -4.8)
= 0 + 0
= 0

Multiclass SVM Loss

$$f(x, W) = Wx$$

$$L_i = \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + 1)^2$$





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How should we choose between W and 2W if they both perform the same on the training data?

Regularization: Beyond Training Error

$$L(W) = \underbrace{\frac{1}{N}\sum_{i=1}^{N} L_i(f(x_i, W), y_i)}$$

Data loss: Model predictions should match training data

Regularization: Beyond Training Error

$$L(W) = \underbrace{\frac{1}{N}\sum_{i=1}^{N} L_i(f(x_i, W), y_i)}_{\text{Data loss}} + \underbrace{\lambda R(W)}_{\text{Regularization}}$$

Data loss: Model predictions should match training data

Regularization: Prevent the model from doing *too* well on training data

Regularization: Beyond Training Error

$$L(W) = \underbrace{\frac{1}{N}\sum_{i=1}^{N} L_i(f(x_i, W), y_i)}_{\text{Data loss}} + \underbrace{\lambda R(W)}_{\text{Regularization}}$$

λ = regularization strength (hyperparameter)

Data loss: Model predictions should match training data

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Regularization: Beyond Training Error

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Data loss: Model predictions should match training data

Regularization: Prevent the model from doing *too* well on training data

Simple examples

L2 regularization: $R(W) = \sum_k \sum_l W_{k,l}^2$

L1 regularization: $R(W) = \sum_k \sum_l |W_{k,l}|$

Elastic net (L1 + L2): $R(W) = \sum_k \sum_l \beta W_{k,l}^2 + |W_{k,l}|$

More complex:

Dropout

Batch normalization

Cutout, Mixup, Stochastic depth, etc...

Regularization: Beyond Training Error

$$L(W) = \underbrace{\frac{1}{N}\sum_{i=1}^{N} L_i(f(x_i, W), y_i)}_{\text{Data loss}} + \underbrace{\lambda R(W)}_{\text{Regularization}}$$

λ = regularization strength (hyperparameter)

Data loss: Model predictions should match training data

Regularization: Prevent the model from doing *too* well on training data

Purpose of Regularization:

- Express preferences in among models beyond "minimize training error"
- Avoid **overfitting**: Prefer simple models that generalize better
- Improve optimization by adding curvature

Regularization: Expressing Preferences

$$x = [1, 1, 1, 1]$$

$$w_1 = [1, 0, 0, 0]$$

$$w_2 = [0.25, 0.25, 0.25, 0.25]$$

$$w_1^T x = w_2^T x = 1$$

L2 Regularization

$$R(W) = \sum_k \sum_l W_{k,l}^2$$

Regularization: Expressing Preferences

$$x = [1, 1, 1, 1]$$

$$w_1 = [1, 0, 0, 0]$$

$$w_2 = \boxed{[0.25, 0.25, 0.25, 0.25]}$$

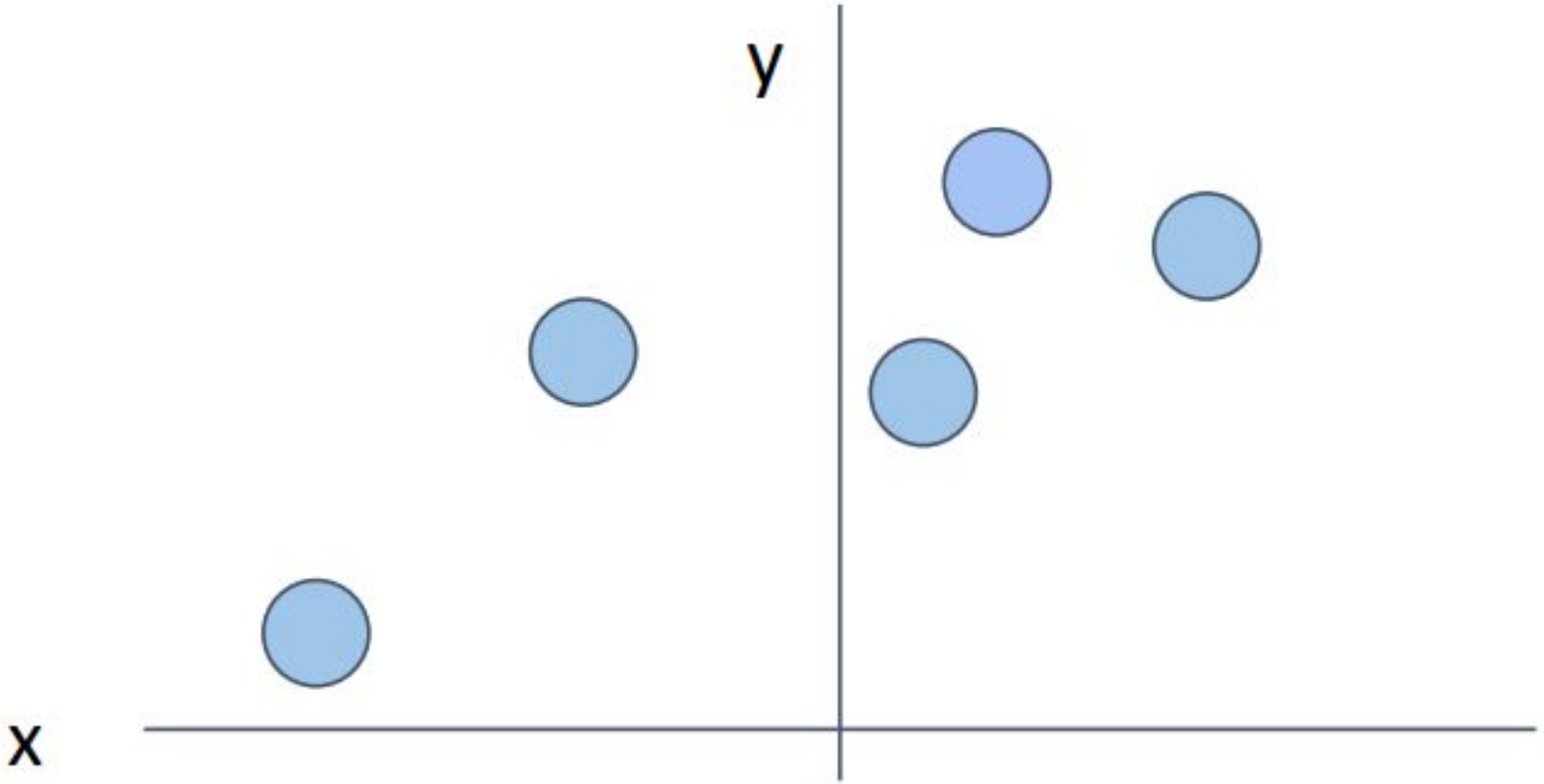
L2 Regularization

$$R(W) = \sum_k \sum_l W_{k,l}^2$$

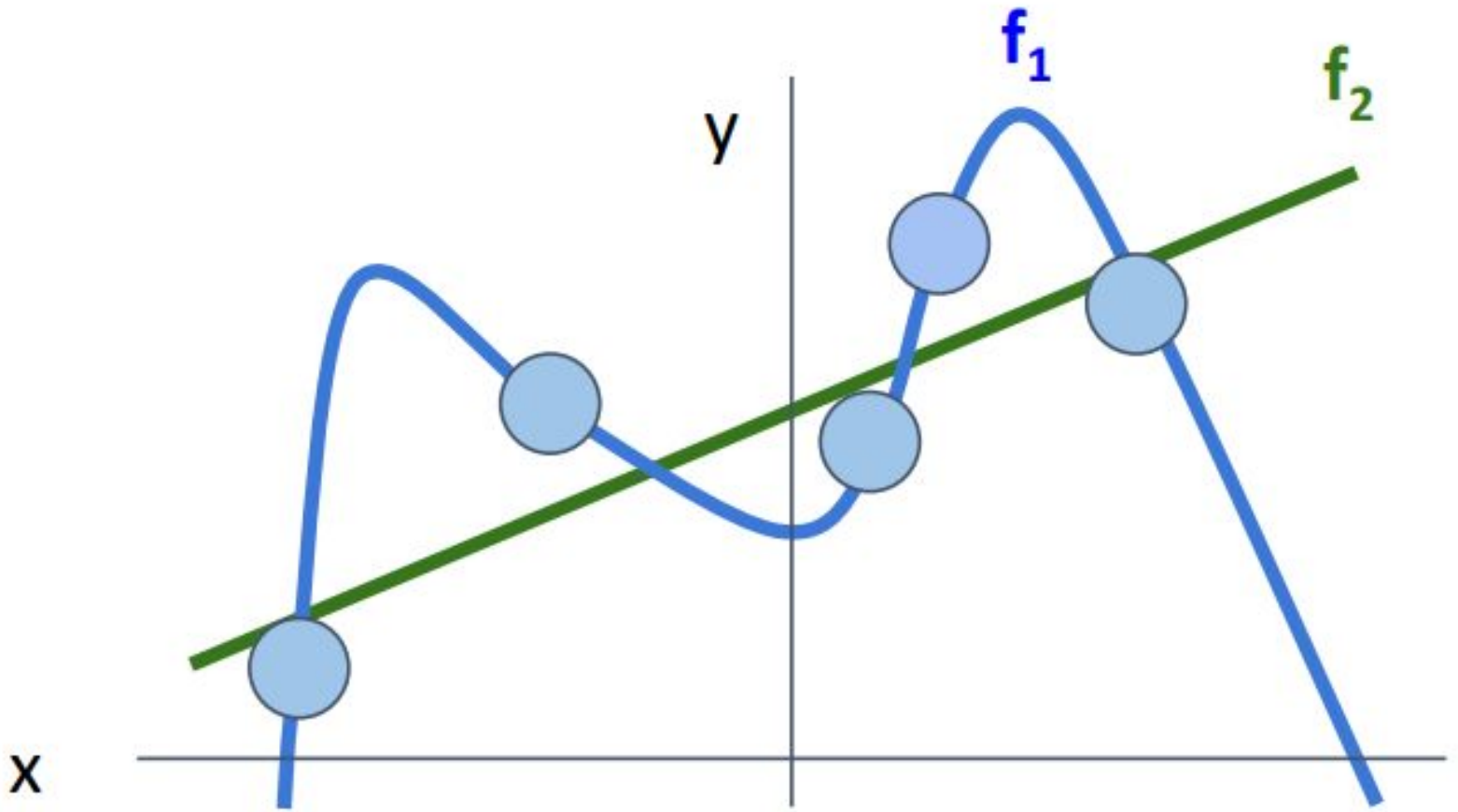
L2 regularization likes to "spread out" the weights

$$w_1^T x = w_2^T x = 1$$

Regularization: Prefer Simpler Models



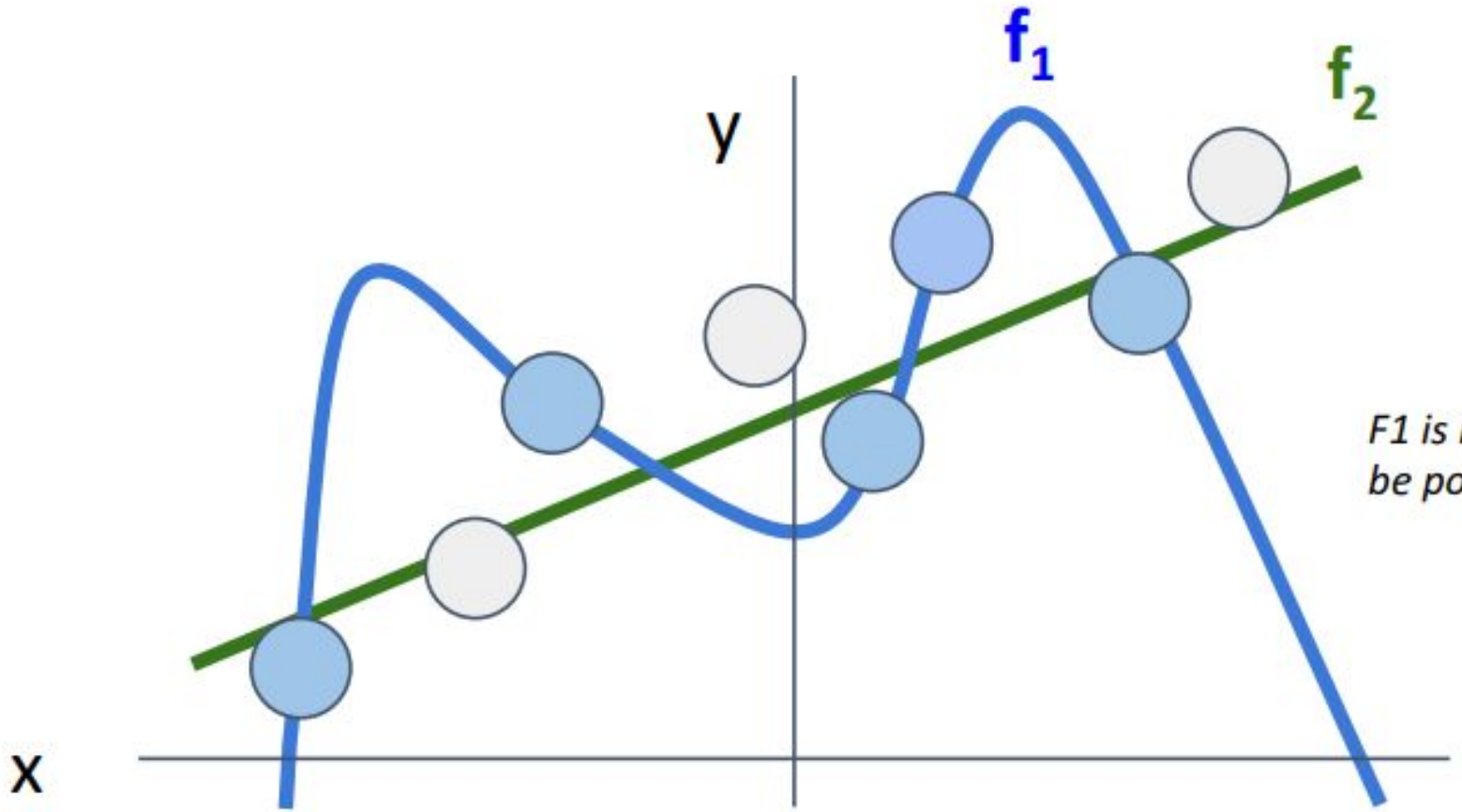
Regularization: Prefer Simpler Models



The model f_1 fits the training data perfectly

The model f_2 has training error, but is simpler

Regularization: Prefer Simpler Models



F1 is not a linear model; could be polynomial regression, etc

Regularization pushes against fitting the data *too* well so we don't fit noise in the data

Part 2 recap and Lab 05 preparation

Dataset loss	Average the per-example losses over the training set.
Regularization	Add a penalty that discourages overly complex parameter choices.
L2 preference	L2 regularization spreads influence across the weights and favors smaller values.
Lab 05	Vectorize the SVM loss, train with minibatches, and evaluate predictions.